



视觉先验 (Prior)

➤ 先验

- 在观测到当前数据（图像、点云等）之前，我们对于未知变量（如深度、光照、物体类别、三维形状）已经拥有的已知假设、约束或概率分布
- 数学上

$$P(\theta | X) \propto P(X | \theta) \cdot P(\theta)$$

- 后验（要的结果） $\propto$ 似然（观测数据提供的证据） $\times$ 先验（提前知道的规律）
- 信息源头上
 - 物理/几何约束：物体不能悬空、两面墙互相垂直
 - 数据统计约束：天空是蓝色的、草是绿色的
 - 人为偏好：自然图像梯度稀疏（平滑）、神经网络权重不要太大（L2正则）.....

规格严格 功夫到家



视觉先验

- 数据分布先验
 - 自监督图像复原
- 先验在损失函数的应用
- 网络结构先验
- 生成式先验

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传统底层视觉先验 - 去噪

- 图像去噪：噪声与图像边缘都是高频，难以区分
 - 平滑性先验（局部恒定）
 - 假设邻域像素值变化平缓。表现为 TV 范数（总变分），惩罚相邻像素的差异，能去噪但会抹平边缘
 - 非局部自相似性（NLM先验）
 - 最强大的传统先验——假设自然图像中同一幅图里存在大量重复的小斑块（Patch）。例如，草地的纹理、砖墙的纹理会在不同位置重复出现。去噪时，用多个相似斑块取平均，噪声会相互抵消（因为噪声均值为零），而真实结构得以保留（BM3D算法就是基于此）
 - 稀疏表示先验
 - 假设图像块在某个过完备字典（如 DCT、小波）下可以用极少数非零系数表示，而噪声无法被稀疏表示，因此可以被滤除

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传统底层视觉先验 - 超分

- 图像超分：一个低分辨率像素对应高分辨率多种排列组合
 - 边缘方向先验
 - 假设图像边缘是各向异性的——沿着边缘方向变化缓慢，垂直边缘方向变化剧烈。插值（如双边滤波）时，只沿边缘方向取点，避免跨边缘取点导致锯齿
 - 梯度轮廓先验（Gradient Profile Prior）
 - 统计大量自然图像发现，边缘横截面的梯度分布服从某种重尾分布（如高斯混合模型）。超分时，强制重建出的高分辨率图像在边缘处的梯度形状符合这种统计规律
 - 跨尺度重复性先验
 - 图像中的小斑块在高分辨率尺度下和低分辨率尺度下会重复出现。即低分辨率图中的某个3×3块，在高分辨率图中能找到对应的更精细的块
 - 自相似性先验（Self-Similarity）
 - 同一张图像内部，尺度缩小后的子块与原图中的子块高度相似。因此，可以利用图像自身的相似结构来“生成”高频细节，无需外部训练集

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Noise2Noise (Nvidia, ICML2018)

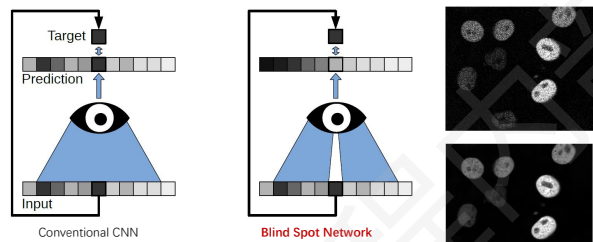
- 使用同场景的另一张图像作为监督

$$\operatorname{argmin}_{\theta} \sum_i L(f_{\theta}(\hat{x}_i), \hat{y}_i)$$

	Gaussian ($\sigma=25$)			Poisson ($\lambda=30$)			Bernoulli ($p=0.5$)		
	clean	noisy	BM3D	clean	noisy	ANSC	clean	noisy	DIP
Kodak	32.50	32.48	31.82	31.52	31.50	29.15	33.01	33.17	30.78
BSD300	31.07	31.06	30.34	30.18	30.16	27.56	31.04	31.16	28.97
Set14	31.31	31.28	30.50	30.07	30.06	28.36	31.51	31.72	30.67
Average	31.63	31.61	30.89	30.59	30.57	28.36	31.85	32.02	30.14

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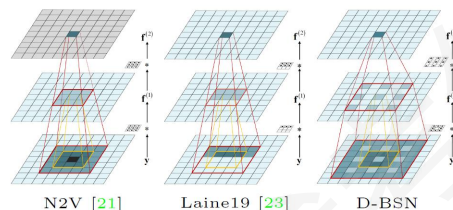
Noise2Void (CVPR 2019)



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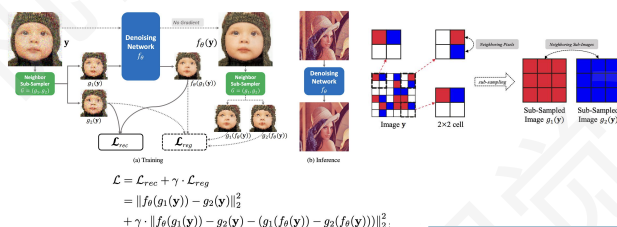
DBSN (ECCV 2020)

盲点网络设计



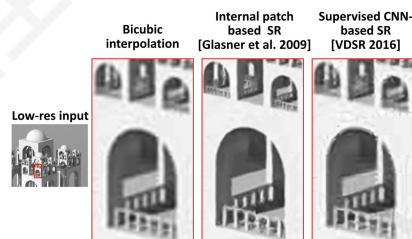
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Neighbor2Neighbor (CVPR 2021)



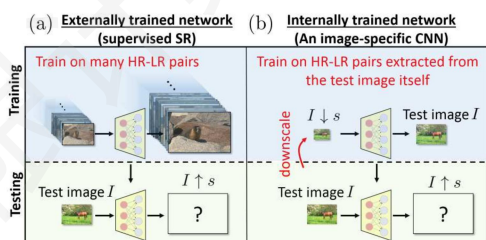
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Zero-shot SR (ZSSR)



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ZSSR



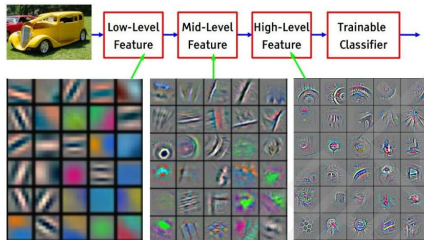
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视觉先验

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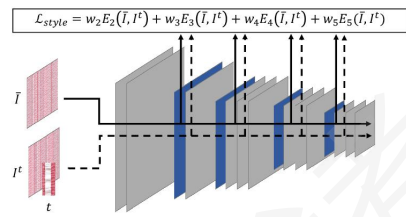
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感知损失 (Perceptual Loss)



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感知损失 (Perceptual Loss)



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全变分损失 (TV Loss)

- 一维连续信号 f 在区间 Ω 的全变分

$$TV_{\Omega}(f) = \int_{\Omega} |\nabla f| dx$$

- 一维离散信号 x 的全变分

$$TV(x) = \sum_n |x_{n+1} - x_n|$$

- 二维连续信号 u 在区间 Ω 的全变分

$$TV_{\Omega}(u) = \int_{\Omega} \sqrt{u_x^2 + u_y^2} dx dy$$

- 二维离散信号 x 的全变分

$$TV(x) = \sum_{i,j} |x_{i+1,j} - x_{i,j}| + |x_{i,j+1} - x_{i,j}|$$

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神经网络的归纳偏置 (Inductive Bias)

- 归纳偏置：算法隐含的“性格偏好”

CNN

- 相比MLP，内置了“空间就近相关”的偏见，认为附近的像素有关系，远处的像素关系弱
- 卷积核共享权重+局部连接，信念是“空间平移等变性”

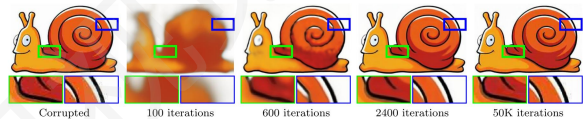
U-Net

- 编码器-解码器结构天然具备低频偏好，偏好生成“平滑、连续、无锯齿”的输出
- U-Net的跳跃连接引入了对高分辨率细节的保真

场景	归纳偏置	数据需求	泛化能力
VIT	弱（没有局部连接）	需要海量数据	数据充足时上限较高
CNN	强（强局部性，平移不变）	少量数据即可	数据少时稳健，数据多时可能不如 VIT

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Deep Image Prior



网络天生更容易学习图像中的低频、有结构的部分

优点：不需要任何训练数据，不需要干净图像，也不需要预训练模型。每一张图像都是独立处理的，非常适合**只有一张退化图像且无法获得干净参考的场景**，比如老照片修复、医学图像去噪。

局限性：需要人为选择停止点，不够自动化；对于每张新图像都需要从头开始训练，计算成本较高（可能需要几分钟到几十分钟）；对于复杂的、结构非常不规则的噪声，效果不如监督方法。

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预训练 GAN 模型

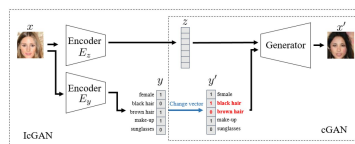


Image Editing (IcGAN)

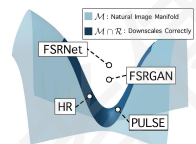
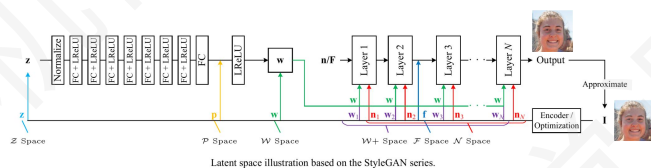


Image Restoration (PULSE)

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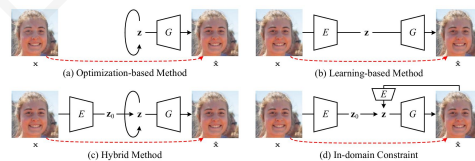
GAN 隐空间 (Latent Spaces) 定义



Latent space illustration based on the StyleGAN series.

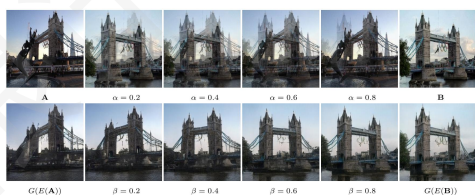
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GAN 反演 (Inversion)



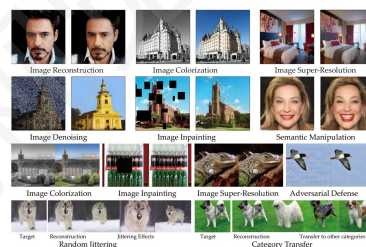
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Application Samples



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Application Samples



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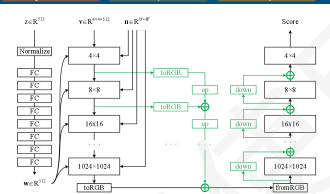
基于预训练 GAN 模型的编辑任务

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Image2StyleGAN

- 分析 StyleGAN 隐空间
- 提出 W^+ 空间
- $w^* = \arg \min_w L_{\text{perceptual}}(G(w), x) + \lambda_{\text{mse}} \|G(w) - x\|_2^2$

Image2StyleGAN: How to Embed Images Into the StyleGAN Latent Space?

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Image2StyleGAN ^{F mins on V100 (32G)}

- Robustness Affine

Mask



Figure 2: Top row: the input images. Bottom row: the embedded results. (a) Standard embedding results. (b) Translation 140 pixels to the right. (c) Translation 140 pixels to the left. (d) Zoom in by 2X. (e) Zoom in by 2X. (f) 90° Rotation. (g) 180° Rotation.

Transformation	$L(\times 10^3)$	$\ w^* - w\ _2$
Translation (Right 140 pixels)	0.782	48.56
Translation (Left 140 pixels)	0.406	44.12
Zoom out (2X)	0.225	38.04
Zoom in (2X)	0.718	40.55
90° Rotation	0.622	47.21
180° Rotation	0.599	42.93

Iterative Embedding



规格严格 功夫到家

Image2StyleGAN

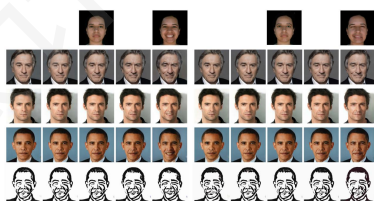
- 应用: Style Mixing (last 9 layers)



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Image2StyleGAN

- 应用: Expression Transfer

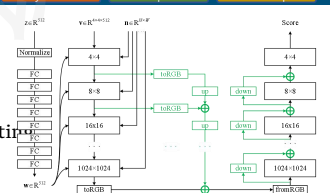


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Image2StyleGAN++

- 引入 $\mathcal{N}$ space
- Optimization objective for editing

Image2StyleGAN++: How to Edit the Embedded Images?

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Image2StyleGAN++

$$L = \lambda_{\text{adv}} \|M_n \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{x})\|_2^2 + \lambda_p L_{\text{percept}}(M_p, G(\mathbf{w}, \mathbf{n}), \mathbf{x}) \\ + \lambda_{\text{adv}} \|(1 - M_n) \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{y})\|_2^2 + \lambda_s L_{\text{style}}(M_s, G(\mathbf{w}, \mathbf{n}), \mathbf{y})$$

重建

$\mathcal{W}^+$ for 5k iters, and $\mathcal{N}$ for 3k iters. PSNR 44~45 dB



规格严格 功夫到家

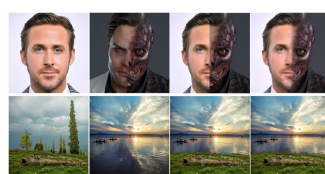


Image2StyleGAN++

$$L = \lambda_{\text{adv}} \|M_n \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{x})\|_2^2 + \lambda_p L_{\text{percept}}(M_p, G(\mathbf{w}, \mathbf{n}), \mathbf{x}) \\ + \lambda_{\text{adv}} \|(1 - M_n) \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{y})\|_2^2 + \lambda_s L_{\text{style}}(M_s, G(\mathbf{w}, \mathbf{n}), \mathbf{y})$$

Image Crossover (语义扩散)

Masked $\mathcal{W}^+$ for 1k iters, and Masked $\mathcal{N}$ for 1k iters.



Algorithm 3: Image Crossover

Input: images $I_1, I_2 \in \mathbb{R}^{n \times m \times 3}$; mask M_{blur}

Output: the embedded code (w_{out}, n_{out})

- 1 $(w^*, n_{ini}) \leftarrow \text{initialize}()$;
- 2 $w_{out} = W_I(M_{blur}, M_{blur}, 1, w^*, n_{ini}, I_1) \\ + W_I(1 - M_{blur}, 1 - M_{blur}, 1, w^*, n_{ini}, I_2);$
- 3 $n_{out} = M_{kn}(M_{blur}, w_{out}, n_{ini}, I_1, I_2);$

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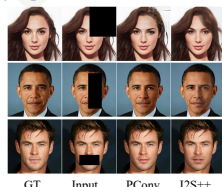


Image2StyleGAN++

$$L = \lambda_{\text{adv}} \|M_n \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{x})\|_2^2 + \lambda_p L_{\text{percept}}(M_p, G(\mathbf{w}, \mathbf{n}), \mathbf{x}) \\ + \lambda_{\text{adv}} \|(1 - M_n) \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{y})\|_2^2 + \lambda_s L_{\text{style}}(M_s, G(\mathbf{w}, \mathbf{n}), \mathbf{y})$$

图像填充

Only optimize 1-9 and 17/18 th w



Algorithm 4: Image Inpainting

Input: image $I_{def} \in \mathbb{R}^{n \times m \times 3}$; masks M, M_{blur}

Output: the embedded code (w_{out}, n_{out})

- 1 $(w_{ini}, n_{ini}) \leftarrow \text{initialize}()$;
- 2 $w_{out} = W_I(1 - M, 1 - M, w_{ini}, w_{ini}, n_{ini}, I_{def});$
- 3 $n_{out} = M_{kn}(1 - M_{blur}, w_{out}, n_{ini}, I_{def}, G(w_{out}));$

规格严格 功夫到家



Image2StyleGAN++

$$L = \lambda_{\text{adv}} \|M_n \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{x})\|_2^2 + \lambda_p L_{\text{percept}}(M_p, G(\mathbf{w}, \mathbf{n}), \mathbf{x}) \\ + \lambda_{\text{adv}} \|(1 - M_n) \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{y})\|_2^2 + \lambda_s L_{\text{style}}(M_s, G(\mathbf{w}, \mathbf{n}), \mathbf{y})$$

局部编辑

Only optimize 4-6 th w



Algorithm 5: Local Edits using Scribble

Input: image $I_{scr} \in \mathbb{R}^{n \times m \times 3}$; masks M_{blur}

Output: the embedded code (w_{out}, n_{out})

- 1 $(w^*, n_{ini}) \leftarrow \text{initialize}()$;
- 2 $w_{out} = W_I(1, 1, w_{ini}, w^*, n_{ini}, I_{scr}) \\ + \lambda \|w^* - w_{out}\|_2;$
- 3 $n_{out} = M_{kn}(M_{blur}, w_{out}, n_{ini}, I_{scr}, G(w_{out}));$

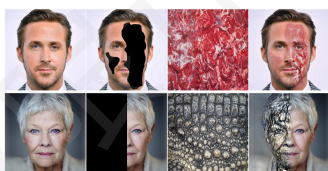
规格严格 功夫到家



Image2StyleGAN++

$$L = \lambda_{\text{adv}} \|M_n \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{x})\|_2^2 + \lambda_p L_{\text{percept}}(M_p, G(\mathbf{w}, \mathbf{n}), \mathbf{x}) \\ + \lambda_{\text{adv}} \|(1 - M_n) \odot (G(\mathbf{w}, \mathbf{n}) - \mathbf{y})\|_2^2 + \lambda_s L_{\text{style}}(M_s, G(\mathbf{w}, \mathbf{n}), \mathbf{y})$$

局部风格迁移



Algorithm 6: Local Style Transfer

Input: images $I_1, I_2 \in \mathbb{R}^{n \times m \times 3}$; masks M_{blur}

Output: the embedded code (w_{out}, n_{out})

- 1 $(w^*, n_{ini}) \leftarrow \text{initialize}()$;
- 2 $w_{out} = W_I(M_{blur}, M_{blur}, 1, w^*, n_{ini}, I_1) \\ + M_{kn}(1 - M_{blur}, w^*, n_{ini}, I_2);$
- 3 $n_{out} = M_{kn}(M_{blur}, w_{out}, n_{ini}, I_1, G(w_{out}));$

规格严格 功夫到家



In-domain GAN Inversion

8 scs

像素级重建 → 语义重建

S 空间 ($\mathcal{W}^+$ 空间)

In-Domain GAN Inversion for Real Image Editing

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ECCV 2020

StyleGAN

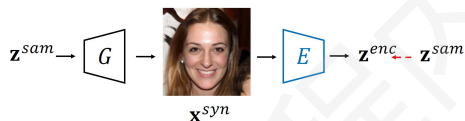
Learn+Opt

 $\mathcal{S}(\mathcal{W}^*)$ Space

In-domain GAN Inversion

像素级重建

$$\min_{\Theta_E} L_E = \|\mathbf{z}^{sam} - E(G(\mathbf{z}^{sam}))\|_2$$



规格严格 功夫到家



ECCV 2020

StyleGAN

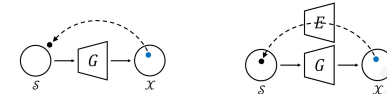
Learn+Opt

 $\mathcal{S}(\mathcal{W}^*)$ Space

In-domain GAN Inversion

语义级重建

- 以模型预测结果 $\mathbf{z}$ 为初始值
- 经过编码器 E 进行优化



$$prev: \mathbf{z}^* = \arg \min_{\mathbf{z}} \|\mathbf{x} - G(\mathbf{z})\|_2 + \lambda_{reg} \|\varphi(\mathbf{x}) - \varphi(G(\mathbf{z}))\|_2$$

$$curr: \mathbf{z}^* = \arg \min_{\mathbf{z}} \|\mathbf{x} - G(\mathbf{z})\|_2 + \lambda_{reg} \|\varphi(\mathbf{x}) - \varphi(G(\mathbf{z}))\|_2 + \lambda_{dom} \|\mathbf{z} - E(\mathbf{z})\|_2$$

规格严格 功夫到家



ECCV 2020

StyleGAN

Learn+Opt

 $\mathcal{S}(\mathcal{W}^*)$ Space

In-domain GAN Inversion

Qualitative comparison Inversion & Editing

Table 1. Quantitative comparison between different inversion methods. For each model, we invert 500 images for evaluation. $\downarrow$ means lower number is better.

Method	Speed	Face			Tower		
		FID $\downarrow$	SWD $\downarrow$	MSE $\downarrow$	FID $\downarrow$	SWD $\downarrow$	MSE $\downarrow$
Traditional Encoder [36]	0.008s	88.48	100.5	0.507	73.02	69.19	0.455
MSE-based Optimization [29]	290s	58.04	20.19	0.026	69.16	55.35	0.068
Domain-Grained Encoder (Ours)	0.015s	82.85	13.02	0.002	40.81	27.13	0.071
In-Domain Inversion (Ours)	8s	42.64	13.44	0.030	44.77	26.44	0.052

Table 2. Quantitative comparison on image interpolation and manipulation between Image2StyleGAN [29] and our in-domain inversion. $\downarrow$ means lower number is better.

Method	Interpolation			Manipulation		
	FID $\downarrow$	SWD $\downarrow$	MSE $\downarrow$	FID $\downarrow$	SWD $\downarrow$	MSE $\downarrow$
MSE-based Optimization [29]	112.09	38.20	121.38	83.69	28.48	113
In-Domain Inversion (Ours)	91.18	33.91	67.22	70.43	17.99	67.92

规格严格 功夫到家



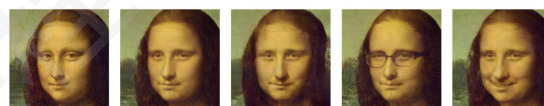
ECCV 2020

StyleGAN

Learn+Opt

 $\mathcal{S}(\mathcal{W}^*)$ Space

In-domain GAN Inversion editing & interpolation



$$\mathbf{x}^{interp} = G(\alpha \mathbf{z}' + (1 - \alpha) \mathbf{z})$$

$$\mathbf{x}^{edit} = G(\mathbf{z} + \alpha \mathbf{n})$$

规格严格 功夫到家



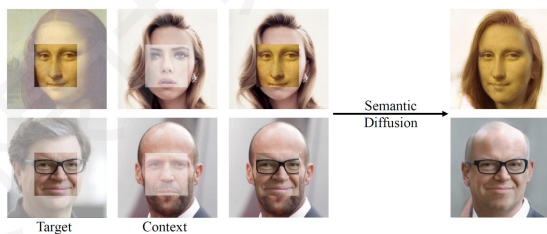
ECCV 2020

StyleGAN

Learn+Opt

 $\mathcal{S}(\mathcal{W}^*)$ Space

In-domain GAN Inversion Semantic Diffusion



$$\mathbf{x}^{diff} = G(E([\mathbf{x}^{outer}, \mathbf{x}^{inner}]))$$

规格严格 功夫到家



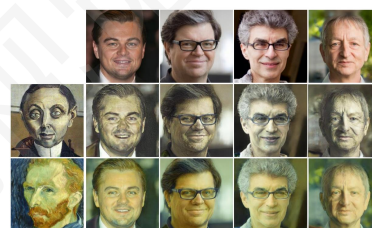
ECCV 2020

StyleGAN

Learn+Opt

 $\mathcal{S}(\mathcal{W}^*)$ Space

In-domain GAN Inversion Style Mixing



$$\mathbf{x}^{mixing} = G([\mathbf{z}'_{coarse}, \mathbf{z}'_{fine}])$$

规格严格 功夫到家

Hair Mapper

- 引入头发掩码 (mask)
- female → male → bald → female

HairMapper: Removing Hair from Portraits Using GANs

Yiqian Wu^{1,2} Yong-Liang Yang³ Xiaogang Jin^{1,2*}

¹State Key Lab of CAD&CG, Zhejiang University

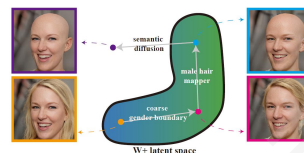
²ZJU-Tencent Game and Intelligent Graphics Innovation Technology Joint Lab

³University of Bath

onethousand@zju.edu.cn, y.yang@cs.bath.ac.uk, jin@cad.zju.edu.cn

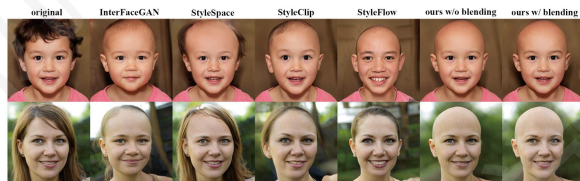
规格严格 功夫到家

Hair Mapper Pipeline



规格严格 功夫到家

Hair Mapper Pipeline



规格严格 功夫到家

预训练 GAN 模型进行图像复原

规格严格 功夫到家

PULSE

5 secs (V100)

PULSE: Self-Supervised Photo Upsampling via Latent Space Exploration of Generative Models

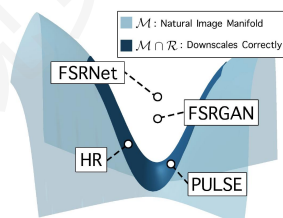
Sachit Menon*, Alexandru Damian*, Shijia Hu, Nikhil Ravi, Cynthia Rudin
Duke University
Durham, NC

{sachit.menon, alexandru.damian, shijia.hu, nikhil.ravi, cynthia.rudin}@duke.edu

规格严格 功夫到家

PULSE

- 预训练 GAN 模型代表的自然图像流形

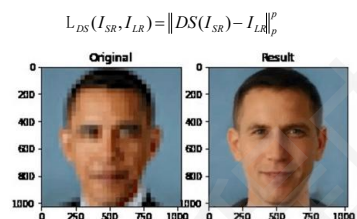


规格严格 功夫到家



PULSE

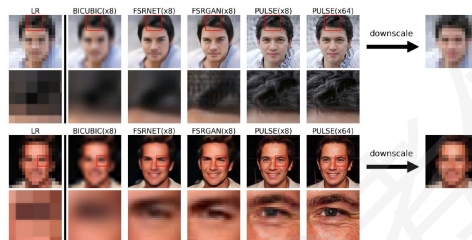
优化目标



规格严格 功夫到家



PULSE



规格严格 功夫到家



pSp

0.105 secs

- Learning-based Encoder $pSp(\mathbf{x}) := G(E(\mathbf{x}) + \bar{\mathbf{w}})$
- LPIPS Loss + $\bar{\mathbf{w}}$ regularization + ArcFace-based ID Loss
- "Invert first, edit later" $\rightarrow$ One stop operation

Encoding in Style: a StyleGAN Encoder for Image-to-Image Translation

Elad Richardson¹ Yuval Alaluf^{1,2} Or Patashnik^{1,2} Yotam Nitzan²
Yaniv Azar¹ Stav Shapiro¹ Daniel Cohen-Or²

¹Penta-AI ²Tel-Aviv University

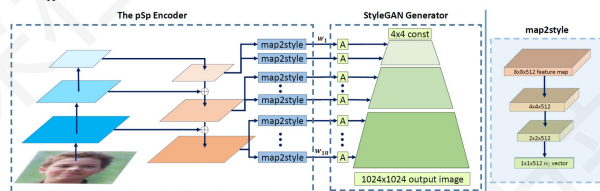
规格严格 功夫到家



pSp

ResNet-50

coarse/medium/fine: 0-2/3-6/7-



规格严格 功夫到家



pSp

损失函数

$$L_2(\mathbf{x}) = \|\mathbf{x} - pSp(\mathbf{x})\|_2, \text{ where } pSp(\mathbf{x}) = G(E(\mathbf{x}) + \bar{\mathbf{w}})$$

LPIPS

$$L_{LPIPS}(\mathbf{x}) = \|F(\mathbf{x}) - F(pSp(\mathbf{x}))\|_2$$

$\bar{\mathbf{w}}$ Regularization

$$L_{reg}(\mathbf{x}) = \|E(\mathbf{x}) - \bar{\mathbf{w}}\|_2$$

ID (R is pre-trained ArcFace Model)

$$L_{id} = 1 - \langle R(\mathbf{x}), R(pSp(\mathbf{x})) \rangle$$

规格严格 功夫到家



pSp

重建

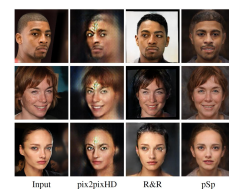
Method	↑ Similarity	↓ LPIPS	↓ MSE	Runtime
Karras et al. [11]	0.77	0.11	0.02	182.1
AI-Art [15]	0.86	0.22	0.15	0.207
DForent [12]	0.18	0.22	0.06	0.032
pSp	0.56	0.17	0.03	0.105

Table 1: Quantitative results for image reconstruction.



Figure 4: Results of pSp for StyleGAN2 generation compared to other methods on CIFAR-100.

Face Frontalization

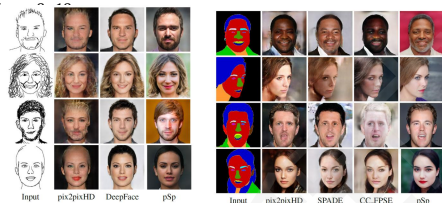


规格严格 功夫到家

pSp

草图/语义图生成图像

$[w]_{inp}^{1-}$



$W =$

规格严格 功夫到家

GPEN

- 同时利用 W^+ 和 $\mathcal{N}$ 空间
- 同时优化生成器

GAN Prior Embedded Network for Blind Face Restoration in the Wild

Tao Yang¹, Peiran Ren¹, Xuansong Xie¹, and Lei Zhang^{1,2*}

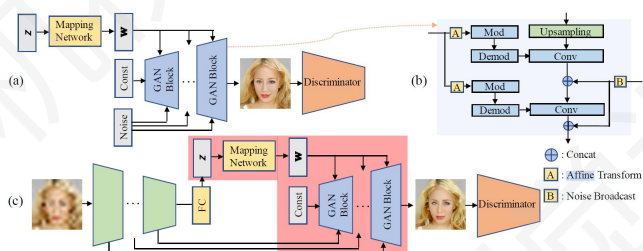
¹DAMO Academy, Alibaba Group

²Department of Computing, The Hong Kong Polytechnic University

yangtao9009@gmail.com, peiran_r@sohu.com, xingtong.xxs@taobao.com, cslzhang@comp.polyu.edu.hk

规格严格 功夫到家

GPEN



规格严格 功夫到家

GPEN

- Learning Objective ($LR_{enc}: LR_{dec}: LR_{dis} = 100: 10: 1$)

$$L_C = \min_G \|X - \tilde{X}\|$$

$$L_A = \min_G \max_{\theta} E_x \log(1 + \exp(-D(G(\tilde{X})))$$

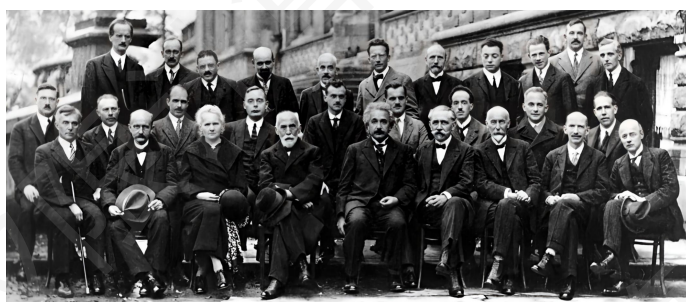
$$L_F = \min_G E_x (\sum_{t=0}^T \|D(X) - D(G(\tilde{X}))\|_2)$$

- Degradation Model

$$I^d = ((I \otimes \mathbf{k}) \downarrow_s + \mathbf{n}_{\sigma})_{JPEQ_q}$$

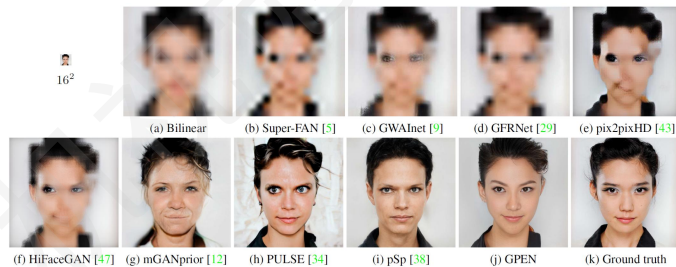
$$s = \{10: 200\}, \sigma = \{0: 25\}, q = \{5: 50\}$$

规格严格 功夫到家



规格严格 功夫到家

GPEN



规格严格 功夫到家

隐空间探索

规格严格 功夫到家

GANSpace

- 无监督GAN隐空间探索

GANSpace: Discovering Interpretable GAN Controls

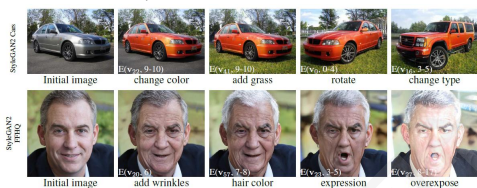
Erik Härkönen^{1,2}Aaron Hertzmann²Jaakko Lehtinen^{1,3}Sylvain Paris²¹Aalto University²Adobe Research³NVIDIA

规格严格 功夫到家

GANSpace

- StyleGAN

- 采样N个随机向量 $z_1, \dots, z_N, w_i = M(z_i)$
- 在W进行主成分分析, $w' = w + Vx$, 找到最显著的变化



规格严格 功夫到家

SeFa

- 隐空间语义方向的解析解

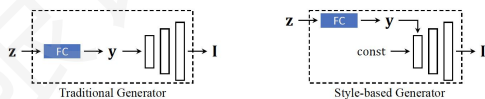
Closed-Form Factorization of Latent Semantics in GANs

Yujun Shen Bolei Zhou
The Chinese University of Hong Kong
{sy116, bzhou}@ie.cuhk.edu.hk

规格严格 功夫到家

SeFa

- 假设
- y的剧烈变化同样会导致I的变化



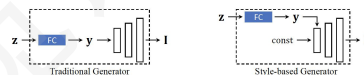
$$y = FC(z) = Az + b.$$

规格严格 功夫到家

SeFa

- Formulation

- Editing on Z space
 $z' = z + \alpha n$, $\text{edit}(G(z)) = G(z') = G(z + \alpha n)$
- Reflection on y
 $\Delta y = FC(z') - FC(z) = (A(z + \alpha n) + b) - (Az + b) = \alpha An$
- Finding n such that αAn is largest
 $n^* = \arg \max_{\|n\|_2=1} \|An\|_2^2$



规格严格 功夫到家

SeFa

解析解推导

假设在Z空间编辑

$$z' = z + \alpha n, \text{ edit}(G(z)) = G(z') = G(z + \alpha n)$$

反映在y上

$$\Delta y = FC(z') - FC(z) = (A(z + \alpha n) + b) - (Az + b) = \alpha A n$$

找到n使得 $\alpha A n$ 最大

$$n^* = \arg \max_{\{n \in \mathbb{B}^d, \|n\|_2=1\}} \|A n\|_2^2$$

规格严格 功夫到家

SeFa

解析解推导

令n为单位向量

$$N^* = \arg \max_{\{n \in \mathbb{B}^d, \|n\|_2=1\}} \sum_{i=1}^k \|A n_i\|_2^2$$

拉格朗日乘法

$$\begin{aligned} N^* &= \arg \max_{\{n \in \mathbb{B}^d\}} \sum_{i=1}^k \|A n_i\|_2^2 - \sum_{i=1}^k \lambda_i (n_i^T n_i - 1) \\ &= \arg \max_{\{n \in \mathbb{B}^d\}} \sum_{i=1}^k (n_i^T A^T A n_i - \lambda_i n_i^T n_i + \lambda_i) \end{aligned}$$

求导

$$2A^T A n_i - 2\lambda_i n_i = 0$$

规格严格 功夫到家

SeFa

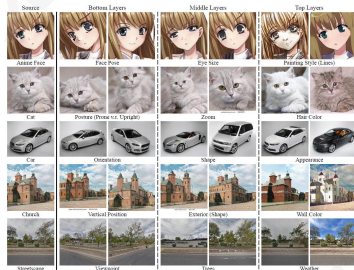


Figure 2. Hierarchical interpretable directions discovered in the style-based generators, i.e., StyleGAN [17] and StyleGAN2 [18]. Among them, the interpretable model is trained with StyleGAN2, while the others are using StyleGAN.

规格严格 功夫到家

SeFa

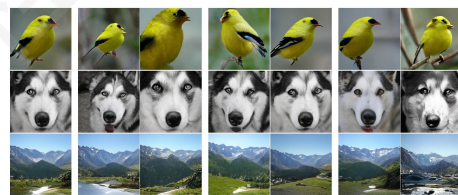


Figure 3. Diverse interpretable directions found in the BigGAN [4], which is conditionally trained on ImageNet [6]. These semantics are further used to manipulate images from different categories.

规格严格 功夫到家

StyleCLIP

通过 CLIP (text) 寻找语义方向

StyleCLIP: Text-Driven Manipulation of StyleGAN Imagery

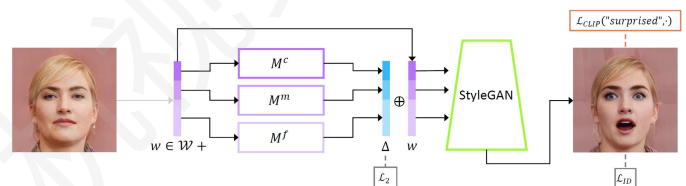
Or Patashnik^{1*} Zongze Wu^{1*} Eli Shechtman[§] Daniel Cohen-Or[†] Dani Lischinski[‡]
[†]Hebrew University of Jerusalem [†]Tel-Aviv University [§]Adobe Research

规格严格 功夫到家

StyleCLIP

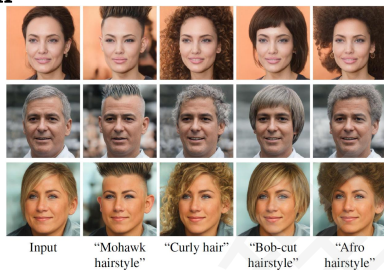
Latent Mapper

$$\begin{aligned} L_{CLIP}(w) &= D_{CLIP}(G(w + M_t(w)), t) \\ L(w) &= L_{CLIP}(w) + \lambda_{L2} \|M_t(w)\|_2 + \lambda_{ID} L_{ID}(w) \end{aligned}$$



规格严格 功夫到家

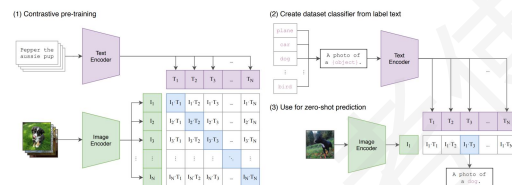
StyleCLIP



规格严格 功夫到家

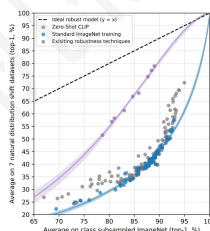
视觉语言预训练模型：CLIP (OpenAI)

➢ 400M (image, text) 成对数据



规格严格 功夫到家

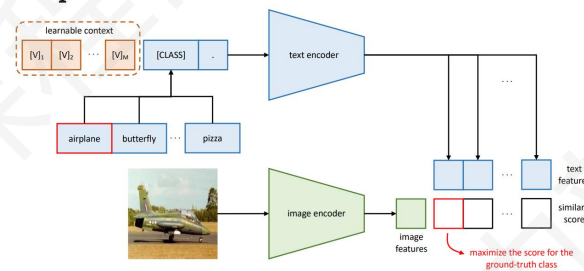
性能



Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet	76.2	76.2	0%
ImageNetV2	64.3	70.1	+5.8%
ImageNet-R	37.7	88.9	+51.2%
ObjectNet	32.6	72.3	+39.7%
ImageNet Sketch	25.2	60.2	+35.0%
ImageNet-A	2.7	77.1	+74.4%

规格严格 功夫到家

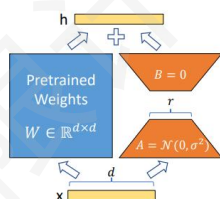
CoOp



Learning to Prompt for Vision-Language Models, IJCV 2022

规格严格 功夫到家

LoRA



$$h = W_0x + \Delta Wx = W_0x + BAx$$

规格严格 功夫到家

The End
THANKS FOR YOUR ATTENTION

规格严格 功夫到家